MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Adult Guideline for Administration of Antimicrobials via Midline Catheter

Purpose

The purpose of this guideline is to provide recommendations for antibiotics that can be infused via a midline catheter, specifically for those agents commonly started during inpatient admission and intended for outpatient antibiotic therapy (OPAT).

Background

Historically, peripherally inserted central catheters (PICC) have been widely used in patients who require long-term central venous access, particularly for those needing courses of outpatient antibiotics.[1,4] Recently, there has been increased interest in the use of midline catheters, which present reduced risk of infection and central venous stenosis when compared to PICCs, and may lower overall IV therapy costs.[2,3,4,25] Typically used for IV medications for up to 14 days, midlines do have some drawbacks, such as increased rates of mechanical complications and mixed rates of associated thrombosis.[1,4] However, a review of 987 articles showed favorable comparisons of midlines against other types of catheters in terms of failure and infection rates.[26]

Because midline tips do not reside in central circulation, they cannot be used for continuous vesicant therapy, parenteral nutrition, or infusates with an osmolarity greater than 900 mOsm/L.[15] There is ongoing debate about the appropriate medications for midline use, The Infusion Nurses Society (INS) guidelines suggest facilities develop their own midline use protocols. [12] This guideline provides recommendations for use of specific antimicrobials planned to be administered as outpatient therapy via midline catheter. We primarily use pH 5-9, and secondarily, we use osmolarity of <900. However, there are antibiotics that are known irritants within these ranges. Thus, we evaluate primary literature for known adverse effects. In the absence of primary literature, we defer to known standard practice across the country, and if other hospitals are using these antibiotics via midline, then we assume the incidence of ADEs is low. It should also be noted that the tables listed in Exhibit B are not comprehensive, and the absence of a drug from the list does not imply that it is either safe or unsafe for use.

Considerations

- Duration of therapy:
  - Recommended duration of therapy via midline is 14 days; however, extended duration of therapy up to 28 days may be considered in certain clinical scenarios. Consider ID consult if an extended duration is required.
- Absolute exclusion criteria:
  - Known hypersensitivity/allergy to approved antibiotics
  - Prior phlebitis or vein injury to planned antibiotic via peripheral administration
  - Known contraindication to midline such as recent thrombosis within 30 days in same limb as planned midline placement, no available vein or decreased venous flow per assessment by VAST, or a vein preservation strategy.
  - Antibiotics requiring continuous infusion – contraindicated due to potential complications from traction on midline catheter[6]
- Relative exclusion criteria:
  - Recent infection or occlusion of midline
  - Age <18 years. Clinicians should reference pediatric literature. This guideline did not examine pediatric midline use.
  - History of thrombosis and hypercoagulability
  - Consideration should be made in terms of patient's anticoagulation status, clot history/timeline/location, and if previously provoked by line placement.

***Providers can further reference the Improve PICC Guidelines for vascular access queries;*
https://www.improvepicc.com/

Exhibit B

Antimicrobials which may be administered via midline [17,18,20-22.24,29,33-37,40-41]		
Antimicrobial	**pH**	**Osmolarity**
Cefazolin	4.5-7.19	270-351
Cefepime	4-6	307
Ceftaroline	5-7	
Cefiderocol	5.1-5.3	572
Ceftazidime	5-8	---
Ceftazidime/avibactam	5.81	495
Ceftolozane/tazobactam	5.9-6.0	500
Ceftriaxone	6.6-6.7	270-423
Daptomycin	4.7-6.8	~323-364
Ertapenem	7.5	---
Meropenem	7.3-8.3	300
Meropenem/vaborbactam	7.9	429-432
Micafungin	5-7	---
Piperacillin/tazobactam	5.5-6.8	600-700
Tigecycline	4-6	240-320

Antimicrobials NOT recommended to be administered via midline (central line preferred for outpatient use, e.g. PICC) [1,5,13,38-41]			
Antimicrobial	**pH**	**Osmolarity**	**Comments**
Acyclovir	10.5-11.6	316	Irritant, vesicant
Amikacin	3.5-5.5	349	Vesicant
Ampicillin	8-10	400	Irritant
Ampicillin/sulbactam	8-10	400	Irritant
Amphotericin B liposomal	5-6	250-270	Irritant
Ganciclovir	11	310	Irritant, vesicant
Gentamicin	3-6	293-320	Irritant, vesicant
Nafcillin	6-9	295-406	Vesicant, case reports of tissue necrosis due to extravasation
Oxacillin	6-7	295-406	Potential vesicant
Sulfamethoxazole/trimethoprim	<5	750-800	Irritant
Tobramycin	3-7	260-320	Vesicant
Vancomycin	2-5	240-290	Irritant, vesicant

It should also be noted that the tables are not comprehensive, and the absence of a drug from the list does not imply that it is either safe or unsafe for use. Requests to add additional antimicrobials to the included list should be sent to medusepolicy@med.umich.edu for consideration.

Administration & Monitoring

- Per Michigan Medicine Nursing Assessment and Care of Venous Access Devices policy Venous Access Devices: Assessment and Care (Venous Access Grid)
- Per Post-Acute Care Services Midline Catheter Care, Policy Stat ID: 6687848 (pending)

Restrictions

VAST approval and placement of midline catheter

Workgroup/Guideline Authors

Inpatient Pharmacy - Michelle Schultz, Adamo Brancaccio
ID Pharmacy - Jerod Nagel
Hospital Medicine - David Paje
Vascular Access Service Team (VAST) - Deanna Skicki
Home Med - Lisa Klein, Maria Hagan, Elizabeth Sayler
Michigan Visiting Nurses - Amy Sweet

Exhibits:

Nursing Assessment and Care of Venous Access Devices
Venous Access Grid
Infection Prevention for Intravenous Peripheral Short Catheters Policy
Nursing Midline Catheter Removal
https://www.improvepicc.com/key-guidelines.html

Last revised: 06/2024	**Antimicrobial Stewardship Subcommittee:** 06/2024	**P&T:** 06/2024

References

	Citation
1	Bahl A, Karabon P, Chu D. Comparison of Venous Thrombosis Complications in Midlines Versus Peripherally Inserted Central Catheters: Are Midlines the Safer Option? Clin Appl Thromb Hemost. 2019 Jan-Dec;25:1076029619839150.
2	Chopra V et al. The Michigan Appropriateness Guide for Intravenous Catheters (MAGIC): Results From a Multispecialty Panel Using the RAND/UCLA Appropriateness Method. Ann Intern Med. 2015 Sep 15;163(6 Suppl):S1-40.
3	Adams DZ et al. The Midline Catheter: A Clinical Review. J Emerg Med. 2016 Sep;51(3):252-8.
4	Cawcutt KA et al. Optimizing vascular-access device decision-making in the era of midline catheters. Infect Control Hosp Epidemiol. 2019 Jun;40(6):674-680.
5	Le A and Patel S. Extravasation of Noncytotoxic Drugs: A Review of the Literature. Ann Pharmacother. 2014 Jul;48(7):870-886. Epub 2014 Apr 8.
6	Dickson et al. A Cluster of Failures of Midline Catheters in a Hospital in the Home Program: A Retrospective Analysis. J Infus Nurs. 2019 Jul/Aug;42(4):203-208.
7	Underwood J et al. Intravenous catheter-related adverse events exceed drug-related adverse events in outpatient parenteral antimicrobial therapy. J Antimicrob Chemother. 2019 Mar 1;74(3):787-790.
8	Seo et al. The Safety of Midline Catheters for Intravenous Therapy at a Large Academic Medical Center. Ann Pharmacother. 2020 Mar;54(3):232-238.
9	Reynolds et al. Management of extravasation injuries: a focused evaluation of noncytotoxic medications. Pharmacotherapy. 2014 Jun;34(6):617-32.
10	Vesely T et al. The diverse and conflicting standards and practices in infusion therapy. Journal of Vascular Access Devices (2002) 7 (3): 9–25.
11	Clark E et al. Reducing risk of harm from extravasation: a 3-tiered evidence-based list of pediatric peripheral intravenous infusates. J Infus Nurs. 2013 Jan-Feb;36(1):37-45.
12	Ryder M et al. Investigation of the role of infusate properties related to midline catheter failure in an ovine model. Am J Health Syst Pharm. 2020 Aug 7;77(16):1336-1346.
13	David V et al. Extravasation of Noncytotoxic Drugs. Ann Pharmacother. 2020 Aug;54(8):804-814.
14	Gorski L, Hagle M, and Bierman S. Intermittently delivered IV medication and pH: reevaluating the evidence. J Infus Nurs. Jan-Feb 2015;38(1):27-46.
15	Infusion Nurses Society. (2021) "Vascular Access Device (VAD) Planning" in Gorski L et al Infusion Therapy Standards of Practice. WolterKluwer. pp 59.

16	Harwood I, Greene L, Kozakowski-Koch J, et al. New peripherally inserted midline catheter: A better alternative for intravenous antibiotic therapy in patients with cystic fibrosis. Pediatric Pulmonology 1992 April 12; 4: 233-239.
17	Ertapenem. Package insert. Merck & Co; 2012
18	Cefepime. Package insert. Hospira; 2012
19	Cefazolin. Package insert. GSK; 2004
20	Ceftriaxone. Package insert. Sandoz; 2013
21	Daptomycin. Package insert. Cubist Pharmaceuticals; 2011
22	Frankenfeld et al. Daptomycin: a comparison of two intravenous formulations. Drug Des Devel Therapy 2018; 12: 1953–1958.
23	Meropenem. Package insert. AstraZeneca; 2016
24	Ceftazidime. Package insert. Teligent Pharma; 2017.
25	Nielsen E, Antonsen L, Mensel C, et al. The efficacy of midline catheters-a prospective, randomized, active-controlled study. Int J Infect Dis. 2021 Jan; 102:220-225.
26	Tripathi S, Kumar S, Kaushik S. The Practice and Complications of Midline Catheters: A Systemic Review. Crit Care Med 2021 Feb 1;49(2): e140-e150
27	Cornely O, Pappas P, Young J, et al. Accumulated safety data of micafungin in therapy and prophylaxis in fungal diseases. Expert Opinion on Drug Safety 2011 Mar;10(2):171-83
28	Cappelletty D, Eiselstein-McKitrick K. The Echinocandins. Pharmacotherapy March 2007; 27(3):369-88.
29	Micafungin. Package insert. Astellas Pharma; 2007
30	Benjamin D, Smith B, Arrieta A, et al. Safety and Pharmacokinetics of Repeat-Dose Micafungin in Young Infants. Clinical Pharmacology & Therapeutics January 2010; 87(1): 93-99.
31	Keller S, Dzintars K, Gorski L, et al. Antimicrobial Agents and Catheter Complications in Outpatient Parenteral Antimicrobial Therapy. Pharmacotherapy 2018; 38(4): 476-481.
32	Price A, Morris T, White H et al. Cutaneous photoxic reaction to IV micafungin in the outpatient setting: A Case Report. Clinical Infection in Practice 2021; 11, 100061.
33	Vabomere. Package insert. Melinta Therapeutics, LLC;2023
34	Pipercillin/tazobactam. Package insert. Melinta Therapeutics, LLC;2023
35	Fetroja. Package insert. Shionogi Inc;2021
36	Tigecycline. Package insert. Wyth;2010
37	Avycaz. Package insert. Allergan USA, Inc;2019
38	Keller SC, Dzintars K, Gorski LA, Williams D, Cosgrove SE. Antimicrobial Agents and Catheter Complications in Outpatient Parenteral Antimicrobial Therapy. Pharmacotherapy. 2018;38(4):476-481. doi:10.1002/phar.2099
39	Sulfamethoxazole and Trimethoprim. Package insert. AR Scientific, Inc.;2013
40	ASHP injectable drug information. (2023)
41	Trissel LA Handbook on Injectable Drugs 11th Edition ASHP Bethesda, MD 2001

Revision History

Date	Revision
04/2023	Added micafungin
06/2024	Revised formatting, duration state, and medication tables

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source

`